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Answer Sheet

1. Why does Information Gain help in better splits?

Information Gain tells **how much uncertainty (entropy) is reduced** after splitting. Higher IG = data becomes more pure (more similar labels). So, we choose the feature that **separates classes best**

2. What happens if two features have the same information gain?

If two features have the same information gain, then we can pick any feature.

3. Why is Linear Regression sensitive to outliers?

Because Linear Regression uses MSE to minimize error. Errors are squared in linear regression. Errors are squared means big errors become very big. Outliers create huge errors.

4. Difference between Correlation and Regression?

Correlation: Measures relationship strength (between -1 to 1)

Regression: Finds equation (line) to predict values